

Arvin Zaker

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Education

Doctor of Medicine (MD)

Aug 2025–Present

University of British Columbia (UBC), Vancouver, BC

Honours Bachelor of Science, Translational and Molecular Medicine

Sep 2019–Apr 2023

University of Ottawa, Ottawa, ON; cumulative grade point average: 9.98/10

Research Experience

Medical Imaging Artificial Intelligence

Feb 2026–Present

University of British Columbia, Vancouver, BC

Supervisor: Dr. Ilker Hacihaliloglu

- Curated 14 datasets and built a Python pipeline to train 1,176 models on 38,357 paired mammograms.
- Externally tested selected models on EMBED dataset to identify AI model performance failures and its association with breast density, race, ethnicity, and histology.

E-Cigarette Flavour Policy Evaluation

Feb 2026–Present

University of British Columbia, Vancouver, BC

Supervisor: Dr. Kevin Duan

- Compared the policy effects of comprehensive e-cigarette flavour bans and specialty-retailer restrictions among 1,516 youth across 5 Canadian provinces using CTNS data and survey-weighted difference-in-differences.

Epigenomics and Chromatin Biology 🧬

Mar 2024–Jan 2026

University of Ottawa, Ottawa, ON

Supervisor: Dr. Arvind Mer

- Analyzed clinical exosomal DNA to identify differentially methylated regions and pathways and integrated multi-omics evidence from 44 studies and databases.
- Built a reproducible CUT&RUN workflow for processing, quality control, annotation, and collaborator use.
- Aggregated and visualized CXXC1 mutational data across the genome and plant homeodomain (PHD).

Text-Mining Feature Selection for Anticancer Drug Response 🧬

Sep 2023–Feb 2024

University of Ottawa, Ottawa, ON

Supervisor: Dr. Arvind Mer

- Devised and benchmarked text-mining feature-selection methods across drug-response prediction models and validation datasets.
- Built reproducible local and high-performance computing workflows and visualizations for model evaluation.

Bacterial Genomics

Jul 2023–Feb 2024

University of Ottawa, Ottawa, ON

Supervisors: Dr. Arvind Mer and Dr. Vanessa M. D'Costa

- Built genomic annotation and phylogenetic-analysis workflows and characterized type IV secretion system clusters across 118 *Acinetobacter* bacterial and plasmid strains.

SMC5/6 Cancer Phenogenomics 🧬

May 2022–Aug 2022

University of Ottawa, Ottawa, ON

Supervisors: Dr. Arvind Mer and Dr. Damien D'Amours

- Analyzed SMC5/6 alterations, genomic instability, transcriptomic changes, cancer phenotypes, and survival in a pan-cancer cohort of more than 65,000 patients across 144 studies.

Optogenetic Neurophysiology

Mar 2022–Sep 2024

University of Ottawa, Ottawa, ON

Supervisors: Dr. Arvind Mer and Dr. Greg Silasi

- Co-developed software to quantify hemodynamic and cortical responses to optogenetic stimulation after stroke *in vivo*.

Patient-Derived Xenograft Drug-Response Modelling

Jan 2022–June 2026

University of Ottawa, Ottawa, ON

Supervisor: Dr. Arvind Mer

- Developed the open-source xeMetron package to calculate 250 measures of patient-derived xenograft chemotherapy response.
- Built machine-learning models to identify biomarkers of drug response and quantify anti-cancer drug response in clinical-trial data.

Bacteriophage Discovery

University of Ottawa, Ottawa, ON

Supervisor: Dr. Adam Rudner

- Isolated, amplified, characterized, and annotated 2 novel bacteriophages, Arzan and Khorshid.

Sep 2021–Apr 2022

Muscle Physiology

University of Ottawa, Ottawa, ON

Supervisor: Dr. Jean-Marc Renaud

- Designed electrophysiology experiments on adenosine receptors and skeletal-muscle fatigue and performed tissue staining, fluorescence imaging, and quantitative image analysis.

Nov 2020–Jul 2022

Peer-Reviewed Publications

- Asare-Werehene M, **Zaker A**, Tripathi S, Communal L, Carmona E, Mes-Masson A-M, Tsang BK, Mer A. Spotlight on nuclear PD-L1 in ovarian cancer chemoresistance: hidden but mighty. *Front Immunol.* 2025;16. doi:10.3389/fimmu.2025.1543529.
- Grégoire S, Grégoire J, Yang Y, Capitani S, Joshi M, Sarvan S, **Zaker A**, Ning Z, Figeys D, Ulrich K, Brunzelle JS, Mer A, Couture J-F. Structural insights into an atypical histone binding mechanism by a PHD finger. *Structure.* 2024;32(9):1498–1506.e4. doi:10.1016/j.str.2024.06.017.
- Wu G, **Zaker A** (co-first author), Ebrahimi A, Tripathi S, Mer AS. Text-mining-based feature selection for anticancer drug response prediction. *Bioinform Adv.* 2024;4(1):vbae047. doi:10.1093/bioadv/vbae047.
- Nasser F, Gaudreau A, Lubega S, **Zaker A**, Xia X, Mer AS, D'Costa VM. Characterization of the diversity of type IV secretion system-encoding plasmids in *Acinetobacter*. *Emerg Microbes Infect.* 2024;13(1):2320929. doi:10.1080/22221751.2024.2320929.
- Roy S, **Zaker A**, Mer A, D'Amours D. Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *NAR Cancer.* 2023;5(3):zcad047. doi:10.1093/narcan/zcad047.

Presentations, Posters, Seminars & Abstracts

- Poster and oral presentations; presenting author.** Roy S, **Zaker A**, Mer A, D'Amours D. Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *Canadian Society for Molecular Biosciences International Conference*; May 2023; Ottawa, ON.
- Oral presentation; presenting author.** **Zaker A**, Mer A. Comparative analysis of drug response metrics in patient-derived xenografts. *4th Annual University of Ottawa Faculty of Medicine Research Day*; Sep 2022; Ottawa, ON. Panel question period; 18 students were selected.
- Abstract and presentation; co-presenter.** Featherstone A, Radar A, Salama A, Tiukuvaara S, Ferri O, **Zaker A**, et al. Give us a millet of your time: dehusking the singleton bacteriophage Arzan. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
- Abstract and presentation; co-presenter.** Friesen R, Algharbi S, **Zaker A**, Mahdi O, Gao R, Ferri O, et al. Investigating nucleotide-binding proteins in bacteriophage JohnDoe. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
- Abstract and presentation; co-presenter.** Cildir Z, Abdisalam S, Alam A, **Zaker A**, Toex N, Chandra A, et al. Prophage liberation! A method to increase the titer of the temperate phages Khorsid and Winseler. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
- Seminar; presenter.** **Zaker A**, Mer A. Comparative analysis of drug response metrics in patient-derived xenografts. *University of Ottawa Translational and Molecular Medicine Seminar*; Apr 2022. Twenty-minute presentation and question period.
- Poster presentation; presenting author.** **Zaker A**, Mer A. Analysis of drug response in patient-derived xenografts. *Faculty of Medicine, University of Ottawa*; Feb 2022; Ottawa, ON.
- Poster and oral presentations; presenting author.** **Zaker A**, McCarthy L, Jung D, Karunakaran G, Rudner A. Discovery and characterization of bacteriophage Arzan. *University of Ottawa Faculty of Medicine*; Dec 2021; Ottawa, ON.
- Abstract and poster presentation; presenting author.** **Zaker A**, McRae C, Renaud JM. Role of adenosine receptors in mouse skeletal-muscle fatigue kinetics. *University of Ottawa Undergraduate Research Opportunity Program Symposium*; Mar 2021.

Leadership & Governance

Research and Innovation Lead	Jul 2026–Present
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MD Student Learning Group, University of British Columbia, Vancouver, BC

- Lead the research and innovation portfolio, connecting peers with learning opportunities and promoting evidence-informed practice, emerging technologies, and student-led scholarship.

Vice-President, External Relations	Jul 2026–Present
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Artificial Intelligence in Medicine Society, University of British Columbia, Vancouver, BC

- Helped launch a student community focused on responsible artificial intelligence in health care and build external partnerships for panels, workshops, and collaborative projects.

Chief Technology Officer	Jun 2026–Present
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Learn Freely, nonprofit organization, Vancouver, BC

- Lead development of a learning management system and workflows for tutoring assignments, progress tracking, project management, and organizational collaboration.

Board Member	Jun 2026–Present
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Inclusion, Diversity and Equity Advisory Committee, Doctors of BC, Vancouver, BC; three-year term

- Selected to advise the Board of Directors on equity, diversity, and inclusion across governance bodies, members, and the medical profession.

Founder and President	Jun 2020–Sep 2024
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Hands on Health Magazine, University of Ottawa, Ottawa, ON

- Founded and led a 20+ member team that published 50+ faculty-reviewed, evidence-based health articles.
- Built editorial, review, and communication workflows that sustained publication during remote operations.

Founder and President	Dec 2020–Apr 2023
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Canadians Sharing Locally and Globally uOttawa, University of Ottawa, Ottawa, ON

- Partnered with a local charity to help raise more than \$100,000 for personal protective equipment and ventilators for rural hospitals in India.
- Led advocacy events and social media outreach on global health inequities and social determinants of health.

Teaching & Mentorship

Founder and Production Designer	Apr 2026–Present
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UBC Integrated Anatomy Atlas, University of British Columbia, Vancouver, BC

- Collaborated with UBC anatomy faculty to develop a comprehensive flashcard deck spanning both preclinical years of the UBC MD curriculum.
- Produced anatomy images and 320 structure-identification questions across 14 laboratory modules in Anki, study-site, and simulated bell-ringer formats.
- Integrated the resource into UBC Elentra, making it available to all UBC medical students.

Premedical Student Mentor	Aug 2025–Present
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Supporting Applications, Supporting Interviews (SASI), Community of Support, Temerty Faculty of Medicine, University of Toronto, Toronto, ON

- Mentored 3 premedical students from groups underrepresented in medicine through one-on-one meetings, individualized application strategy and feedback on interview and written submissions.

Science, Technology, Engineering, and Mathematics Tutor	Sep 2023–Jul 2025
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Oxford Learning Oak Ridges and Inspiration Learning, Toronto, ON

- Tutored 20+ students from grades 1–12 and university across biology, chemistry, physics, and mathematics using individualized learning plans.

Peer Tutor and Mentor	Sep 2018–Present
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- Guided 50+ middle-school and high-school students in science and mathematics and supported transitions to university through academic guidance and peer mentorship.

Community Engagement

First-Year Representative	Sep 2025–Sep 2026
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Persian Medical Society, University of British Columbia, Vancouver, BC

- Help create and promote peer-support, mentorship, and cultural events for Persian-identifying students and allies, strengthening community connection and inclusive representation within the medical school.

- Administrative Receptionist** **Sep 2022–Jun 2023**
Chadha Medical Clinic, Ottawa, ON
- Coordinated appointments, referrals, and screenings in a primary-care practice serving approximately 5,000 patients and managed 60+ patient contacts weekly.
- Volunteer Coordinator and Co-Chair** **Sep 2020–Sep 2024**
Best Buddies uOttawa, University of Ottawa, Ottawa, ON
- Recruited and paired 50+ student volunteers with people with intellectual and developmental disabilities.
 - Led accessible events and revised recruitment to address audiovisual and language barriers identified through feedback.

Honours, Scholarships & Awards

- Florence E. Heighway Summer Research Award** **Jun 2026–Aug 2026**
Faculty of Medicine Summer Student Research Program, University of British Columbia, Vancouver, BC
- Awarded 8 weeks of funding for a mammography artificial-intelligence failure-analysis project.
- Third Place, IFHP NavAID Challenge** **Jun 2026**
University of British Columbia, Vancouver, BC
- Collaborated with 2 medical students to develop a 75-minute refugee-health case with 4 decision points and 7 interprofessional objectives.
- Undergraduate Student Research Award (USRA)** **May 2023–Aug 2023**
Natural Sciences and Engineering Research Council of Canada (NSERC), University of Ottawa, Ottawa, ON
- Awarded \$7,500 for a 3-month research project based on academic merit and research-proposal quality; one of 88 awards at the University of Ottawa.
- Translational and Molecular Medicine Summer Training Award** **May 2022–Aug 2022**
University of Ottawa, Ottawa, ON
- Awarded \$8,900 for a 3-month supervised research project based on academic merit and research-proposal quality.
- Work in Biomedical Research Scholarship** **May 2022–Aug 2022**
University of Ottawa, Ottawa, ON
- Awarded \$3,500 for a 3-month research project through a competition with a success rate below 50%.
- Undergraduate Research Opportunity Program Award** **Dec 2020**
University of Ottawa, Ottawa, ON
- Awarded \$1,000 for a 4-month supervised research project based on academic merit and research-proposal quality.
- Dean's Honour List and Merit Scholarship** **Sep 2019–Apr 2023**
University of Ottawa, Ottawa, ON

Projects & Interests

- Explainable Heart-Disease Risk Modelling**  **Mar 2026**
National Canadian Medical Datathon, Vancouver, BC
- Collaborated with 5 colleagues to integrate 920 records from 4 University of California, Irvine heart-disease cohorts and develop a regularized logistic-regression model.
 - Built CLEAR, an R Shiny demonstration tool providing explainable and individualized risk estimates using SHapley Additive exPlanations and plain-language feature summaries.
- Clinical Bioinformatics Reproducibility Project**  **Mar 2023**
- Reproduced the bioinformatics analysis reported by Viñas et al. (2020; PMID: 32662516) and produced a documented report, presentation, and explanatory visualizations.
- Systems Administration and Research Computing** **Jan 2022–Present**
- Built and maintain a 16-terabyte Unraid server for centralized family storage, automated backups, data recovery, and Docker-based web services.
 - Built and administer 3 Linux workstations for machine learning, and high-performance computing.